2.6.5 Update

This is arguably the most interesting of heaps' public methods, even if it's not always directly provided in implementations[12].

When we change an element, its priority could stay unchanged, and in that case, we don't need any further action. But it could also become lower or higher. If an element’s priority becomes higher, we need to check that it doesn't violate the third invariant for its parents; in case it becomes lower, it could violate the same invariant for its children.

In the former situation, we need to bubble up the modified element until we find a higher-priority ancestor, or we reach the root.

In the latter we, instead, push it down until all its children are lower-priority, or we reach a leaf.

Listing 2.6 The Update method

```
function update(oldValue, newPriority)
  index ←
pairs.find(oldValue)
     #1
  if index > 0
then
         #2
    oldPriority ← pairs[index].priority
    pairs[index] ← Pair(oldValue, newPriority)
    if (newPriority < oldPriority)
then                                          #3
```